High-Voltage 2-Wire Current-Output Latch Hall Switch IC

1. Features

- AEC-Q100 qualified
- ISO26262 ASIL B
- 2-Wire Current-Output Latch Hall Switch IC
  - 2-Wire Output Interface
  - Open and short circuit diagnosis
- 2 selectable low current levels:1.2mA/6mA
- Wide voltage range: 4V to 40V
- Wide operating temperature range:
  - -40℃ to 150℃
- Reverse battery protection: -28V
- CRC check protection
- Thermal shutdown protection
- High EMC/ESD immunity
- Package:
  - TO-92S (UA)
  - SOT23-3L (SO)

2. Applications

- Automotive and industrial applications
- Motor speed and rotation direction
- Motor status monitoring
- Limit detection
- Position detection

3. Description

SC25896 is manufactured using BCD process. It is an automotive-grade high-voltage 2-wire current-output latch Hall-effect sensor. Developed in accordance with ISO 26262 standard, the device supports ASIL-B functional safety level. Its low-current operating mode offers two selectable states: 1.2 mA and 6 mA. The 2-wire current-output interface not only reduces wiring harness cost, but also facilitates open-circuit and short-circuit diagnostics.

SC25896 is designed specifically for automotive applications and complies with the AEC-Q100 standard. It supports a wide operating voltage range from 4 V to 40 V, making it well-suited for automotive systems. Meanwhile, the device features high electromagnetic compatibility (EMC) performance and high reliability.

SC25896 adopts dynamic offset cancellation and temperature-compensation technologies to reduce offset caused by process variation, package stress and thermal stress, delivering stable sensitivity. The device integrates internal circuit modules including voltage regulator, Hall array, amplifier circuit, low-pass filter, hysteresis comparator and output-current control circuit.

SC25896 is available in 3-pin through-hole TO-92S (UA) and 3-pin surface-mount SOT23-3L (SO) packages. SC25896 uses 100% lead-free matte tin-plated leads.



Fig.1 Package Outline

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4. Terminal Configuration

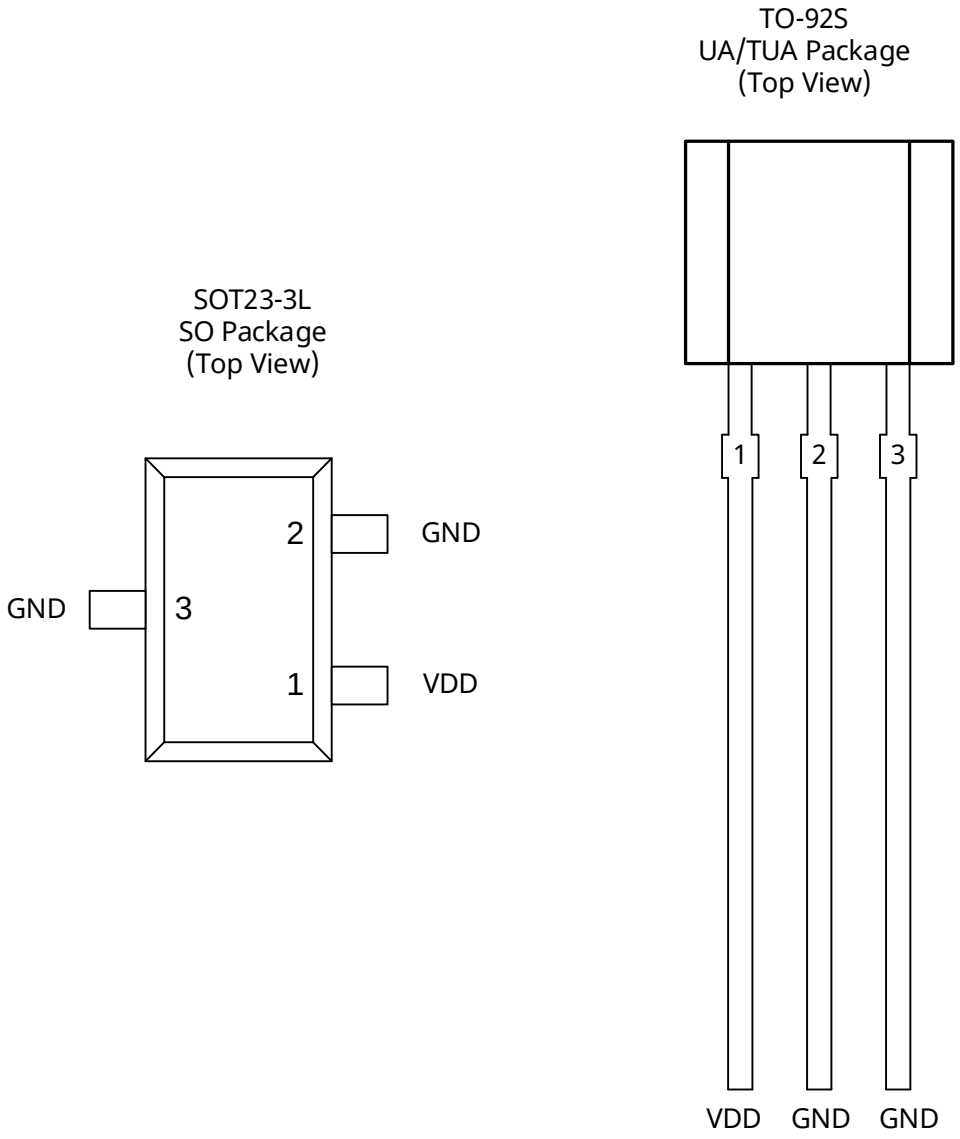


Fig.2 Pin Definition Diagram

Terminal			Type	Description
Name	SO	UA		
VDD	1	1	Power	4V to 40V power supply
GND	2	2	Ground	Ground terminal
GND	3	3	Ground	Ground terminal

5. Ordering Information

Ordering Information	Polarity	I_{DDL}(mA)	I_{DDH}(mA)	B_{OP}(mT)	B_{RP}(mT)	Package	Packing	Quantity
SC25896UA-NLC-BK-Q	North	1.2	14	-8.0	8.0	TO-92S	Bulk	1000/bag
SC25896SO-SHA-TR-Q	South	6	14	2.0	-2.0	SOT23-3L	Tape & Reel	3000/reel
SC25896UA-SHA-BK-Q	South	6	14	2.0	-2.0	TO-92S	Bulk	1000/bag
SC25896SO-SHC-TR-Q	South	6	14	8.0	-8.0	SOT23-3L	Tape & Reel	3000/reel
SC25896UA-SHC-BK-Q	South	6	14	8.0	-8.0	TO-92S	Bulk	1000/bag

Ordering Information Format

SC25896SO-SHA-BK-Q

Product Grade
Q: Automotive Product

Pack Designation
TR: Tape & Reel
BK: Bulk

Magnetic
A : BOP: 20Gs, BRP: -20Gs
C : BOP: 80Gs, BRP: -80Gs

Low-Current Level
L : IDDL:1.2mA
H : IDDL:6mA

Sensing Polarity
S : South-Pole Sensing
N : North-Pole Sensing

Package Designation
SO : SOT23-3L
UA : TO-92S

Device Family
SC25896 : High-Voltage 2-Wire Current-Output Latch Hall Switch IC

6. Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted) (1)

Symbol	Parameter	Test Conditions	Min.	Max.	Units
V_{DD}	Power supply voltage		-28	60	V
I_{DD}	Supply current		0	40	mA
$I_{DD\ REV}$	Reverse supply current		-20	-	mA
T_A	Operating ambient temperature		-40	150	℃
T_J	Maximum junction temperature	For 168h max	-55	165	℃
T_{STG}	Storage temperature		-65	175	℃

Note :
(1) Stresses above those listed here may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods may affect device reliability

7. ESD Protection

Symbol	Parameter	Test Conditions	Min.	Max.	Units
V_{ESD_HBM}	HBM	According to: standard AEC-Q100-002 HBM	-8	+8	kV
V_{ESD_CDM}	CDM	According to: standard AEC-Q100-011 CDM	-2	+2	kV

8. Thermal Characteristics

Symbol	Parameter	Test Conditions	Rating	Units
$R_{\Theta ja}$	UA Package thermal resistance	Single-layer PCB, with copper limited to solder pads	200(1)	℃/W
$R_{\Theta ja}$	SO Package thermal resistance	Single-layer PCB, with copper limited to solder pads	300(1)	℃/W

Note:
(1) Maximum voltage must be adjusted for power dissipation and junction temperature, see Thermal Characteristics

9. Operating Characteristics

9.1. Electrical Characteristics

over operating free-air temperature range V_{DD}=12V (unless otherwise noted)

Symbol	Parameter	Test Conditions	Min.	Typ.(1)	Max.	Units
V_{DD}	Operating voltage	$T_J < T_{J(Max.)}$	4.0	12	40	V
I_{DDL}	Operating supply current	V_{DD}=4V to 40 V, T_A=25℃	0.8	1.2	1.6	mA
			5.0	6.0	6.9	mA
I_{DDH}	Operating supply current	V_{DD}=4V to 40 V, T_A=25℃	12	14	17	mA
$UVLO_H$	Reverse Supply current	B>B_{OP}+2.0mT, V_{DD} Rising From 1.5V	-	1.95	-	V
$UVLO_L$	Safe Mode Supply Current	B>B_{OP}+2.0mT, V_{DD} Decreasing From 2.5V	-	1.85	-	V
$UVLO_{HYS}$	Under Voltage Lockout High	$UVLO_H$- $UVLO_L$	-	100	-	mV
t_{on}	Under Voltage Lockout Low	V_{DD}>5V	-	170	300	μs
t_d	Output delay time	V_{DD} =12V, Rsense=100Ω, CBYP=100nF	-	25	40	μs
Sr	Output Slew Rate	V_{DD} =12V, Rsense=100Ω, CBYP=100nF	1.0	3.0	5.0	μs

Note:

(1) Typical values are defined at T_A = +25°C and V_{DD}=12V

(2) Maximum voltage must be adjusted for power dissipation and junction temperature, see Thermal Characteristics

9.2. Magnetic Characteristics

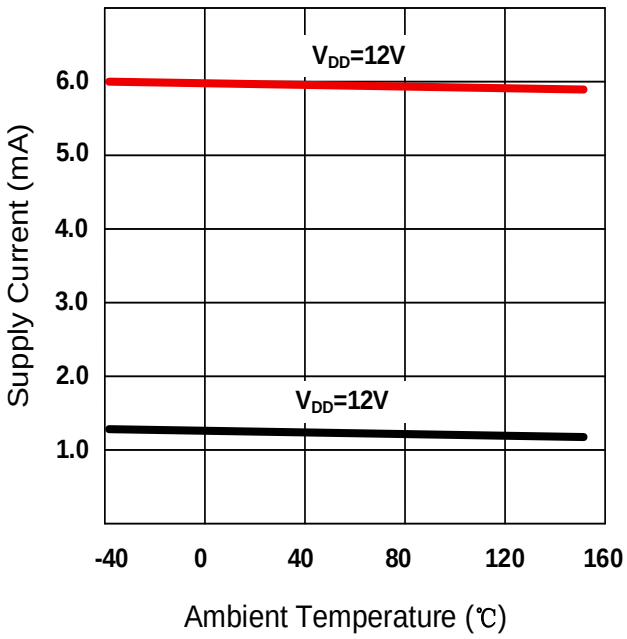
over operating free-air temperature range, V_{DD}=12V (unless otherwise noted)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Units
f_{BW}	BW		20	-	-	kHz
SC25896-XXA 2.0(1) / -2.0 mT(2)						
B_{OP}	Operating point	T_A=25℃	1.0	2.0	3.0	mT
B_{RP}	Release point		-3.0	-2.0	-1.0	mT
B_{HYS}	Hysteresis		-	4.0	-	mT
B_O	Magnetic offset	T_A=25℃, B_O=(B_{OP}+B_{RP})/2	-2.0	0	2.0	mT
SC25896-XXC 8.0 / -8.0 mT						
B_{OP}	Operating point	T_A=25℃	6.0	8.0	10.0	mT
B_{RP}	Release point		-10.0	-8.0	-6.0	mT
B_{HYS}	Hysteresis		-	16.0	-	mT
B_O	Magnetic offset	T_A=25℃, B_O=(B_{OP}+B_{RP})/2	-2.0	0	2.0	mT

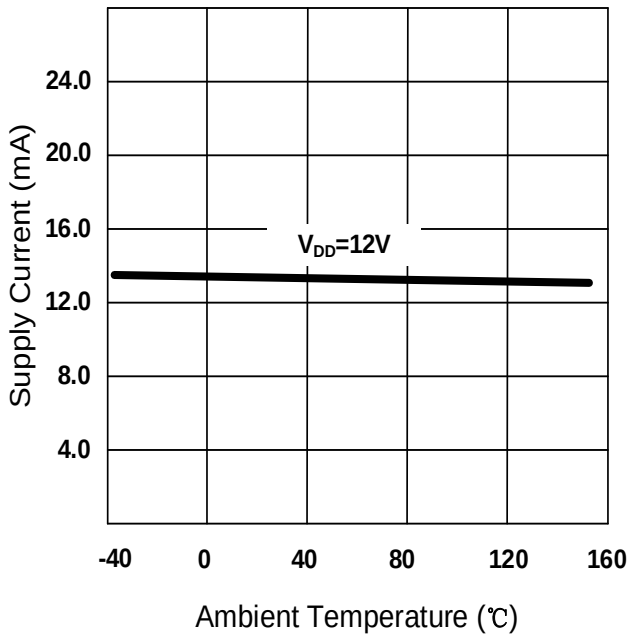
Note:
(1) Magnetic flux density, B is indicated as a negative value for North-polarity magnetic fields, and as a positive value for South-polarity magnetic fields
(2) 1mT=10Gs

10. Characteristic Curves

I_{DDL} vs T_A

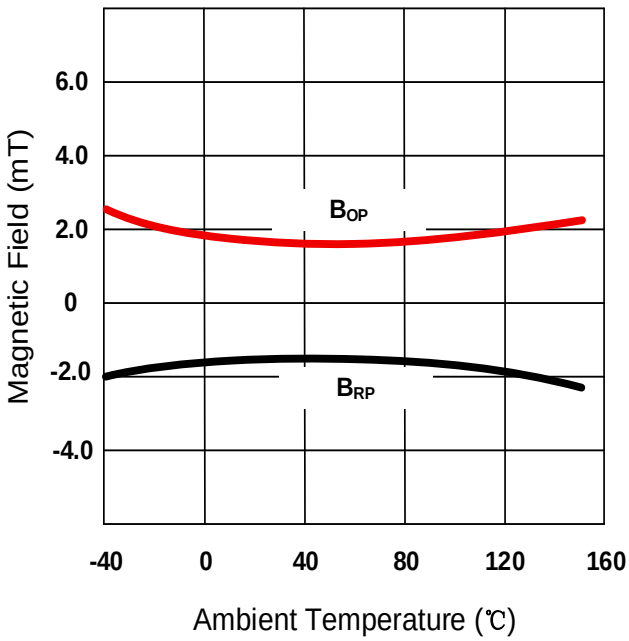


I_{DDH} vs T_A



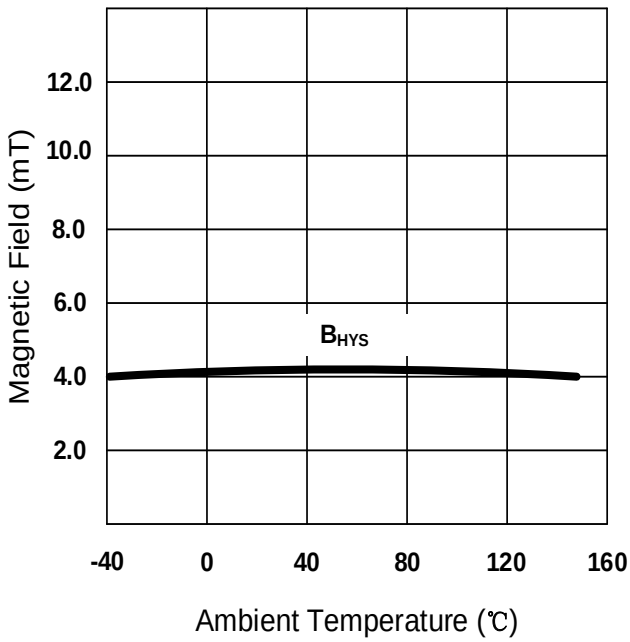
B_{OP} and B_{RP} vs T_A

SC25896-XXA



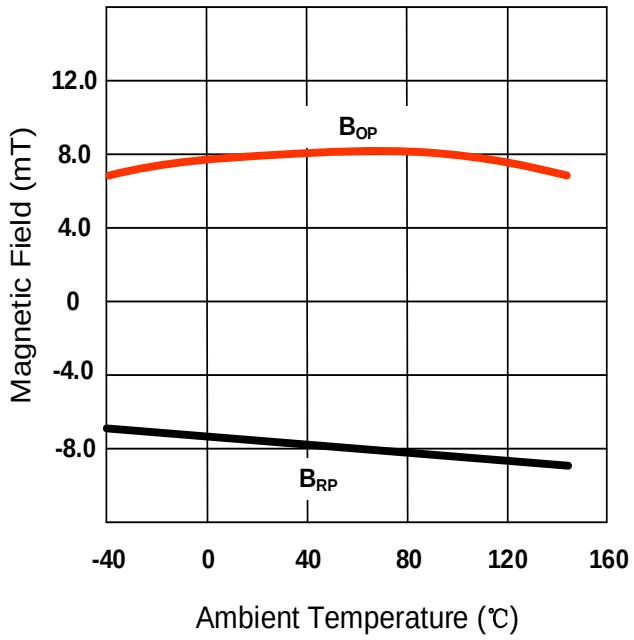
B_{HYS} vs T_A

SC25896-XXA



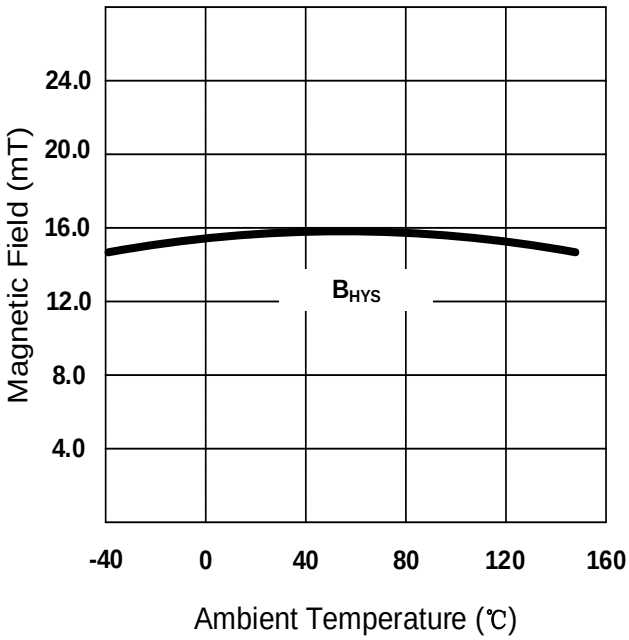
B_{OP} and B_{RP} vs T_A

SC25896-XXC



B_{HYS} vs T_A

SC25896-XXC



11. Block Diagram

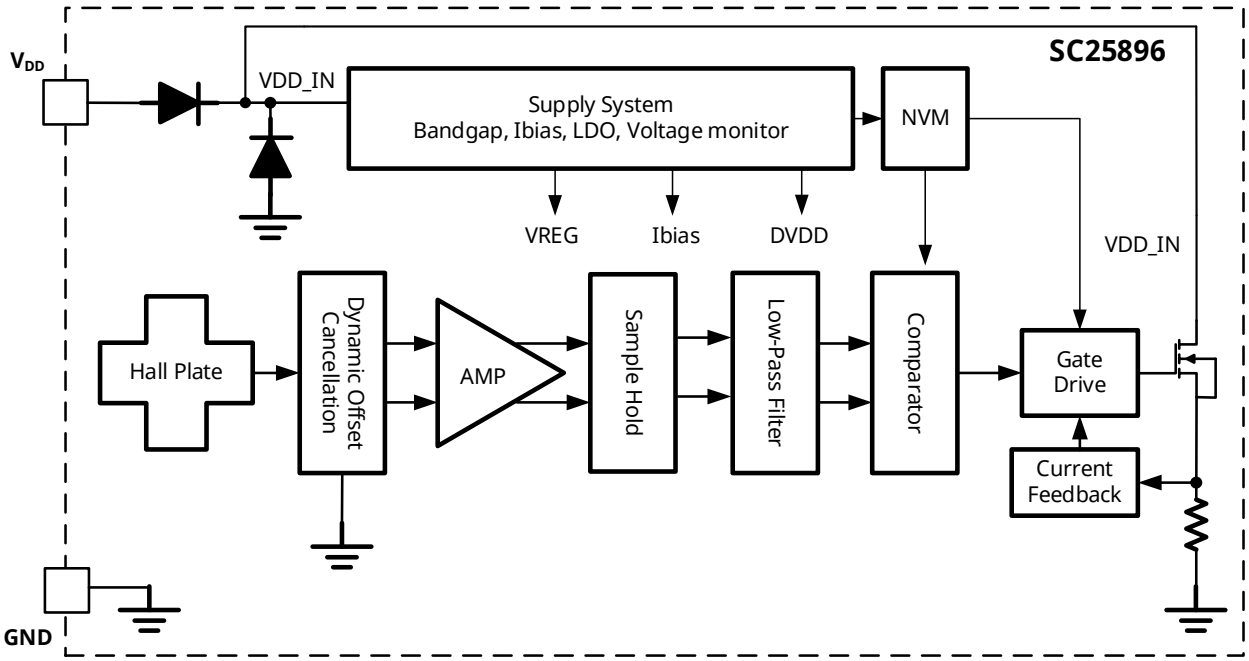


Fig. 3 Functional block diagram

12. Function Description

SC25896 is a high-voltage 2-wire current-output Hall-effect latch sensor adopting high-frequency chopper stabilization technology, which delivers excellent stability of magnetic switching thresholds. The device outputs a valid detection signal when the sensed magnetic flux density exceeds the preset threshold. Two low-current operating levels (1.2 mA and 6 mA) are selectable for flexible application configuration. Optimized for general magnetic sensing scenarios, the device operates reliably across a supply voltage range of 4 V to 40 V and is immune to damage under a reverse voltage of -28 V.

For the SC25896 series, when the magnetic-field strength perpendicular to the Hall element rises to the absolute value of magnetic operating point B_{OP}, the operating current is I_{DDL} (ON). When the magnetic-field strength falls to the absolute value of magnetic release point B_{RP}, the operating current is I_{DDH} (OFF). The difference between the magnetic operating point and magnetic release point is the switching-point hysteresis B_{HYS}. This internal hysteresis enables the device to work reliably under external mechanical disturbances and noise.

12.1. Magnetic Field Direction Definition

A positive magnetic field is defined as the magnetic South pole facing the marked side of the package.

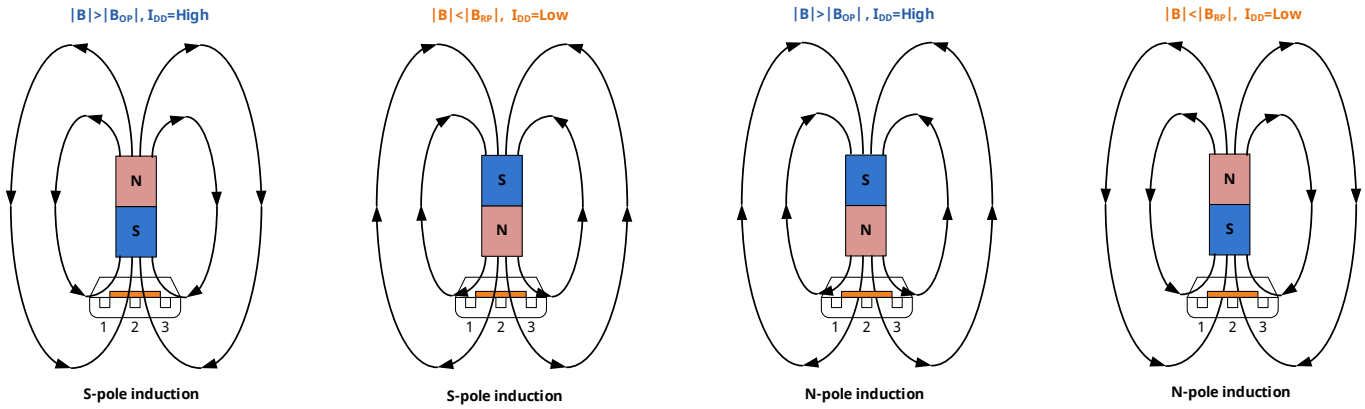


Fig.4 Magnetic Field Direction Definition

12.2. Transfer Function

Powering-on the device in the hysteresis region, less than BOP and higher than BRP, allows an indeterminate output state. The correct state is attained after the first excursion beyond BOP or BRP. If the field strength is greater than BOP, then the output is pulled low. If the field strength is less than BRP, the output is released.

BOP—magnetic threshold for activation of the device output, turning in ON (low) state.

BRP—magnetic threshold for release of the device output, turning in OFF (high) state.

$B_{HYS} = B_{OP} - B_{RP}$

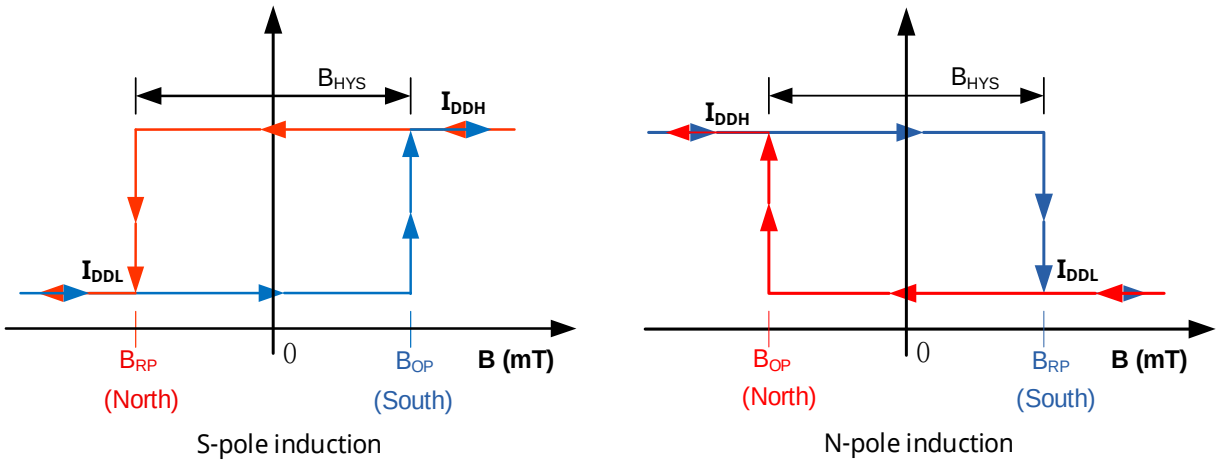


Fig.5 Transfer Curve

12.3. Diagnostic Features

When the SC25896 is powered normally, the operating current has only two states: I_{DDH} or I_{DDL}. Any state with a current value outside this range can be regarded as a fault condition. In the event of a short circuit, the operating current will exceed I_{DDH}; in the event of an open circuit, the operating current will be lower than I_{DDL}. In this way, the connection status between the ECU and the sensor can be easily detected.

13. Typical Application

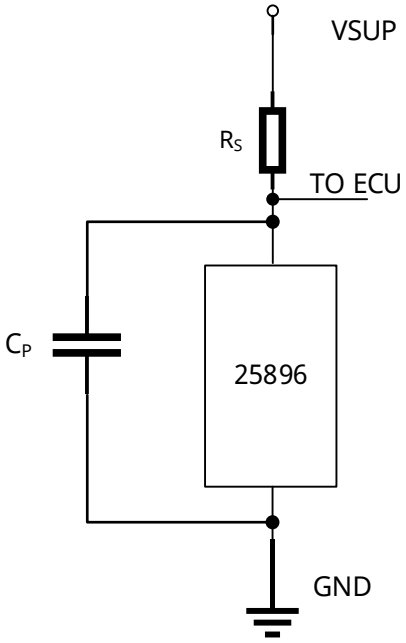


Figure A. High-Side Current-Sensing

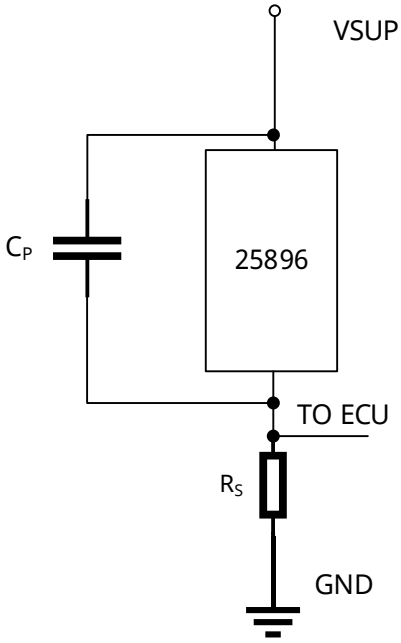


Figure B. Low-Side Current-Sensing

Fig.6 Typical Application Circuit

SC25896 integrates an internal voltage regulator, enabling operation over a wide supply-voltage range.

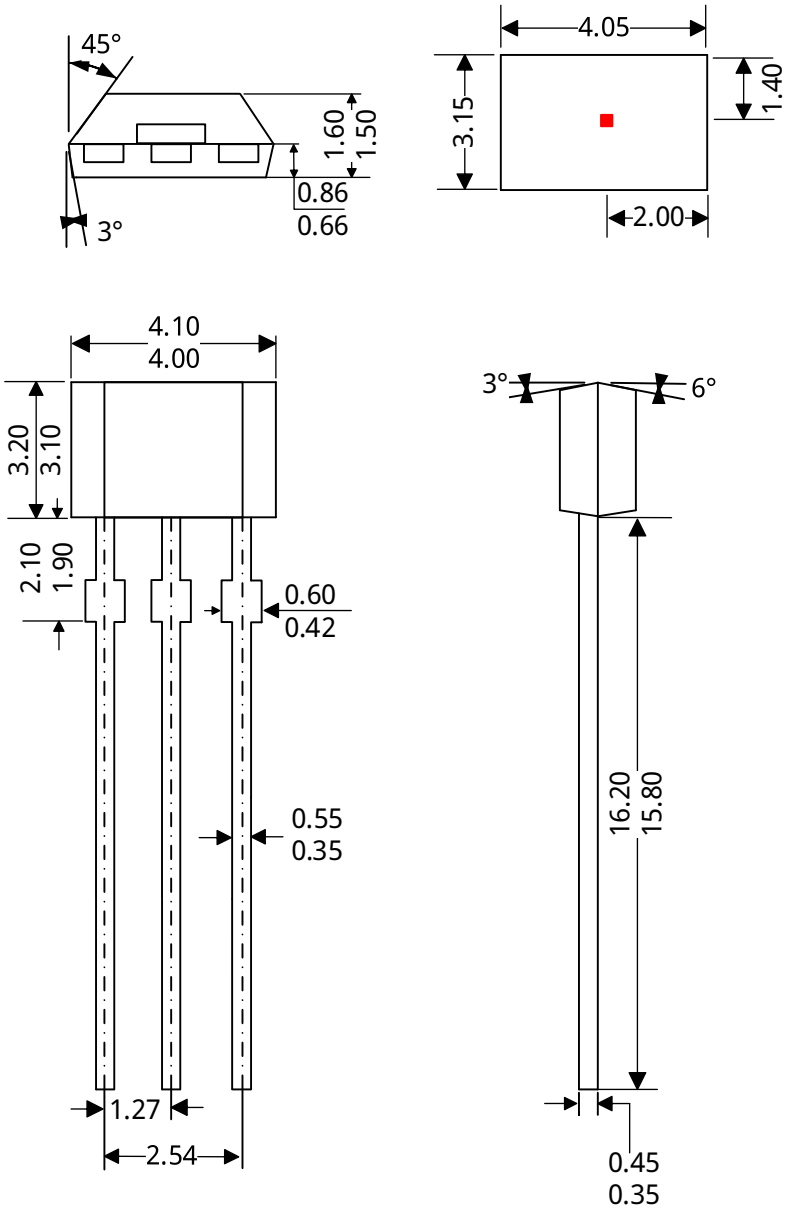
As shown in Figure B above:

For the low-side current-sensing solution with Rs = 360 Ω, the high-level and low-level voltages to ECU are 5.04 V and 0.432 V respectively, which is suitable for 5.0 V applications.

For the low-side current-sensing solution with Rs = 820 Ω, the high-level and low-level voltages to ECU are 11.48 V and 0.984 V respectively, which is suitable for 12.0 V applications.

14. Package Information UA

TO-92S Package Outline Dimensions

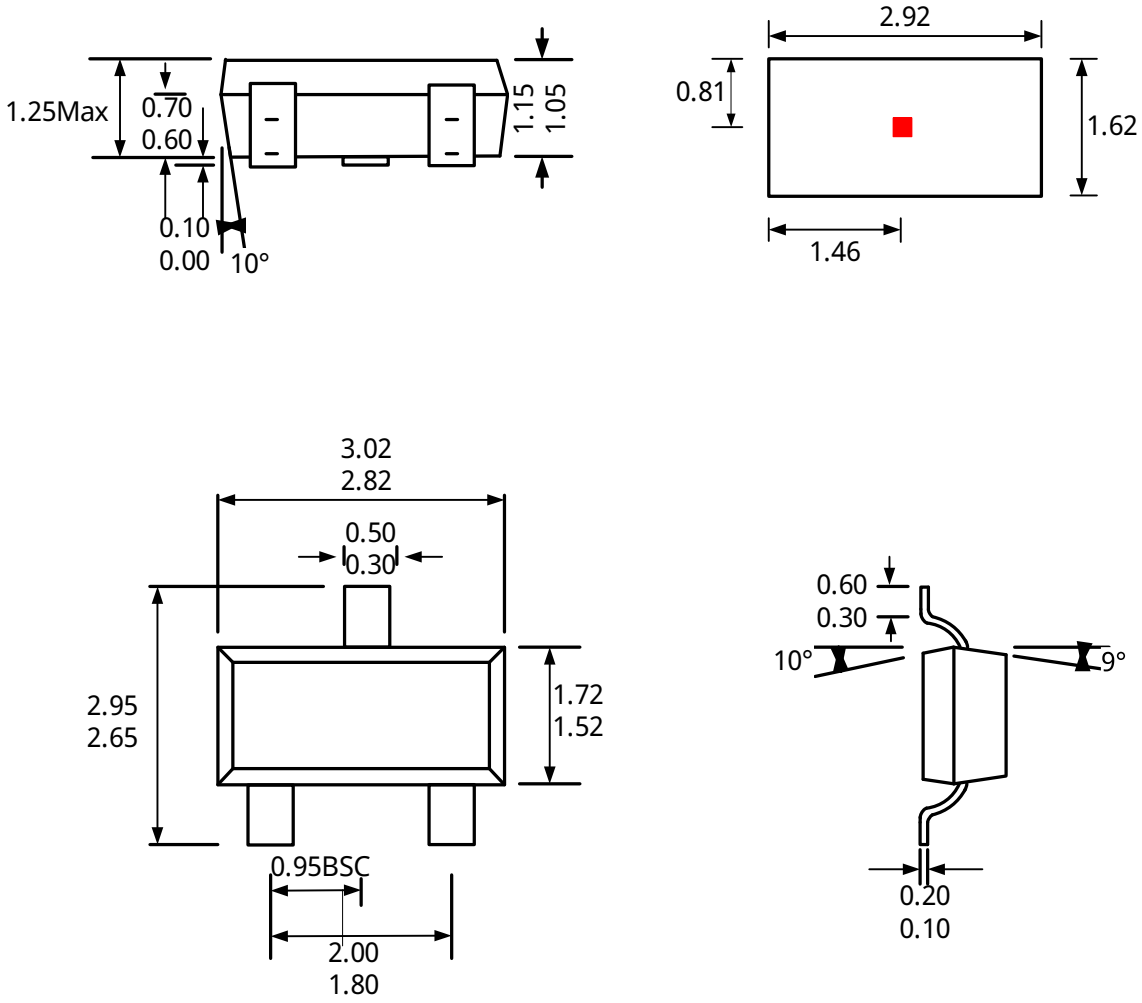


Notes:
(1)All unit in mm.
(2)Dimension does not include mold flash , protrusions or gate burrs.
(3)Allowable dambar protrusion shall be in excess at maximum material condition.

If no tolerance is specified , the dimension shall be theoretical reference value and shall not represent the exact dimension for actual measurement.

15. Package Information SO

SOT23-3L Package Outline Dimensions

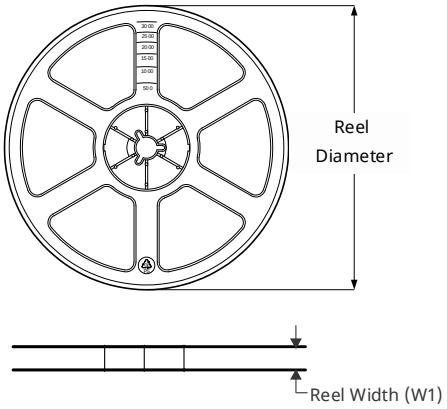


Notes:
(1)All unit in mm.
(2)Dimension does not include mold flash , protrusions or gate burrs.
(3)Allowable dambar protrusion shall be in excess at maximum material condition.

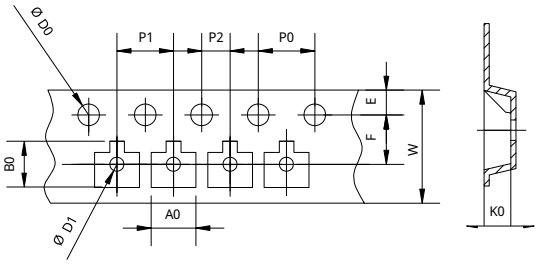
If no tolerance is specified , the dimension shall be theoretical reference value and shall not represent the exact dimension for actual measurement.

16. Tape&Reel Information

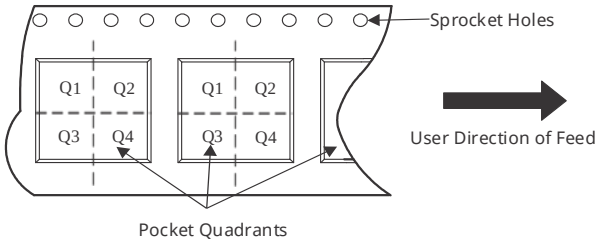
TAPE REEL DIMENSIONS



TAPE DIMENSIONS



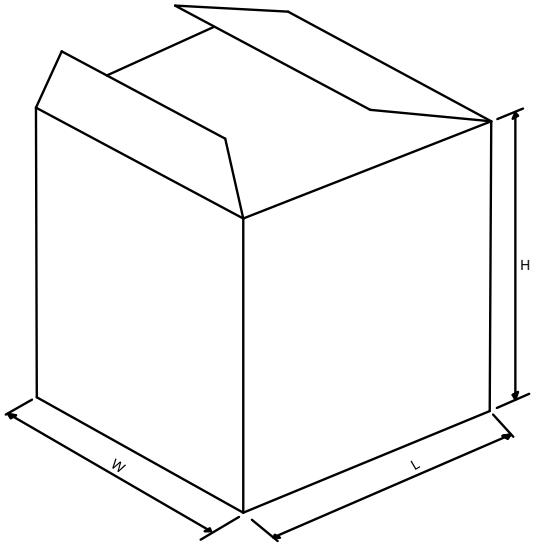
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Package Type	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	P0 (mm)	P1 (mm)	P2 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	W (mm)	Pin1 Quadrant
SOT23-3L	3000	180	8.4	4.00	4.00	2.00	3.18	3.28	1.32	8.00	Q3

17. Tape&Reel Box Dimensions



*All dimensions are nominal

Package Type	SPQ	Length (mm)	Width (mm)	Height (mm)
SOT23-3L	3000ea*10tape	210	210	210

18. Revision History

Revision	Date	Description
Rev.E0.1	2025-08-06	Preliminary Datasheet
Rev.V1.0	2026-02-04	Official version release
Rev.V1.1	2026-08-01	Updated ordering information table

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